

Question	Marking details	Marks available					
		AO1	AO2	AO3	Total	Maths	Prac
7	Starch • Polymer of α-glucose • Composed of amylose and amylopectin • Amylose contains only 1,4 glycosidic bonds • Forms a helical structure • Amylopectin contains 1,4 and 1,6 glycosidic bonds • Forms a branched structure Triglycerides/lipids • Composed of glycerol and three fatty acids • Joined together by ester bonds • Saturated fatty acids contain only C-C single bonds • Unsaturated fatty acids contain at least one C=C double bond • Properties of triglyceride dependent upon the fatty acids they contain Functions in the seed • Starch and triglycerides are insoluble so are osmotically inert • Starch's helical/branched structure makes the molecule compact • Hydrolysis provides glucose readily • Required for respiration/produce ATP • Triglycerides also have a compact structure • Triglycerides have many high energy bonds/ provide approximately twice the quantity of energy than starch	6	3				

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				7-9 marks Indicative content of this level is... - Detailed description of the structure of starch - Detailed description of the structure of lipids/triglycerides - Detailed explanation of how these structures and properties relate to their function in the seed <i>The candidate constructs an articulate, integrated account, correctly linking relevant points, such as those in the indicative content, which shows sequential reasoning. The answer fully addresses the question with no irrelevant inclusions or significant omissions. The candidate uses scientific conventions and vocabulary appropriately and accurately.</i> 4-6 marks Indicative content of this level is... Any two from: - Description of the structure of starch - Description of the structure of lipids/triglycerides - Explanation of how these structures and properties relate to their function in the seed <i>The candidate constructs an account correctly linking some relevant points, such as those in the indicative content, showing some reasoning. The answer addresses the question with some omissions. The candidate usually uses scientific conventions and vocabulary appropriately and accurately.</i>						

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
				1-3 marks Indicative content of this level is... • Brief description of the structure of starch or • Brief description of the structure of lipids/tryglycerides or • Reference to how these structures and properties relate to their function in the seed <i>The candidate makes some relevant points, such as those in the indicative content, showing limited reasoning. The answer addresses the question with significant omissions. The candidate has limited use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give a relevant answer worthy of credit.</i>						
				Question 7 total	6	3	0	9	0	0